Output Sensitivity Modification for Nonlinear Model Predictive Control

Trym Arve Lund Gabrielsen, Lars Struen Imsland, Dinesh Krishnamoorthy

***Abstract*— We present the *Output Sensitivity Modification (OSM)* strategy for dynamic optimization and a corresponding Nonlinear Model Predictive Controller (NMPC): *OSM-NMPC*. The OSM strategy redefines the solutions of a dynamic optimization problem, and characterizes them by a set of *halting conditions*. These conditions are derived from a conceptual *Modified-Gradient Sequential Quadratic Programming (MSQP)* algorithm, which applies a modification to the model input-output sensitivity matrix at each step of an SQP algorithm applied to the dynamic optimization problem. Previous work has used sensitivity modifications in linear MPC to incorporate structural preferences such as input-output pairing/decoupling. These methods are limited to the linear case, and only assess performance and constraint violation with respect to the modified model. The OSM-NMPC scheme addresses this by extending the method to the nonlinear case, and evaluates feasibility using the true model. To illustrate the efficacy of OSM-NMPC, we apply the proposed method to a benchmark, unstable, exothermic, jacketed *Continuous Stirred Tank Reactor (CSTR)* process in a numerical case study, where we decouple the inlet flow rate from the tank temperature. The study also demonstrates that the OSM-NMPC scheme respects the system constraints with respect to the unmodified system model, which leads to automatic 'reactivation' of the unwanted couplings—without extra logic—if necessary to respect system constraints.**

***Index Terms*— Continuous Stirred Tank Reactor, Decoupling, Nonlinear Model Predictive Control, Sensitivity Modification, Sequential Quadratic Programming**

I. Introduction

It is well established that the input-output sensitivity matrix in multivariable control is central in determining the achievable performance of a controller, and its robustness and numerical reliability [1]–[5]. In turn, analysis of its structure gives a foundation for design of controllers that appropriately handle input-output interaction. To this end, measures such as Relative Gain Array, singular values, and directionality measures are used to assess robustness properties and controllability limitations, and to guide control design accordingly. These concerns have led to several strategies for loop-pairing and decentralized control structures [6], [7].

This work is funded by the "ExtruTeC" project of the Research Council of Norway (Project Number: 313919), in partnership with Hydro Extruded Solutions AS, Cybernetica AS, SINTEF Industry, and Norwegian University of Science and Technology, Department of Engineering Cybernetics. Corresponding author T.A.L. Gabrielsen.

All authors are at the Department of Engineering Cybernetics (ITK), Norwegian University of Science and Technology (NTNU), O.S. Bragstads plass 2D, 7034 Trondheim. (email: tagabrie@ntnu.no)

In the context of linear Model Predictive Control (MPC), structural analysis has resulted in methods that have been developed to limit controllers to only using certain structural interactions between inputs and outputs, even when the underlying system does not have these limitations. Such methods include Gain Conditioning, orthogonalization techniques, and decoupling techniques via elimination of weak or uncertain input-output channels [8]–[11]. These are all aimed at improving conditioning of the sensitivity matrix or preventing the controller from exploiting unwanted or numerically fragile input-output couplings.

From an engineering perspective, input-output pairing decisions arise, for example, from the need to alleviate wear on actuators and/or to accommodate operator acceptance [12]. This need has led to both research on advanced methods for extending actuator lifetime (*reliability*) [13], and simpler, more pragmatic approaches such as *MV-CV elimination/decoupling* for MPC, which is a common industrial practice [12], [14]. The latter are employed when developing linearized models, and their effects then persist into all aspects of the controller, such as design, performance, feasibility and robustness considerations.

Some MPC-based strategies address structural design of input-output coupling without modifying the model. These methods include cost function shaping, regularization, and move suppression to encourage the desired behavior [15], [16]. It is usually not straightforward to express the desired behavior exactly in terms of a scalar function, and such methods are therefore often inaccurate with respect to the sought behavior.

In summary, current methods are predominantly linear, lacking the explicit handling of nonlinear dynamics, and are either unable to assess feasibility with respect to unmodified model predictions, or do not accurately represent the desired behavior.

Contribution: In this paper, we address this gap by proposing the *Output Sensitivity Modification* (OSM) strategy for dynamic optimization problems, which can incorporate sensitivity modification strategies directly into nonlinear dynamic optimization, while preserving the underlying model, thus preserving feasibility and objective evaluation with respect to the true model. The proposed OSM strategy is then used to define the OSM-NMPC scheme. This results in a controller that respects the control structure considerations, such as pairing decisions, yet exploits knowledge about the nonlinear model to evaluate constraint satisfaction and performance.

Specifically, we formalize the notion of *halting conditions*, which redefine the 'solution' of a dynamic optimization

problem under the OSM strategy and are derived from a conceptual Modified-gradient Sequential Quadratic Programming (MSQP) algorithm. The OSM-NMPC feedback law is subsequently defined by the OSM strategy, i.e. by the *halting conditions* rather than the *optimum* of the dynamic optimization problem.

The paper is outlined as follows: Section II introduces the conceptual MSQP algorithm and halting conditions. Building on these results, in Section III, we define the OSM strategy for dynamic optimization problems and the corresponding OSM-NMPC scheme. A numerical case study that highlights the OSM-NMPC's ability to decouple the system, yet maintain coupled feasibility, follows in Section IV. The paper ends with concluding remarks in Section V.

II. Modified-Gradient SQP Methods

In Section III, we present a strategy for dynamic optimization and a corresponding NMPC scheme, based on a modified version of the KKT conditions: *Halting Conditions*. In this section, we first formalize and derive these halting conditions from a conceptual modification of the Sequential Quadratic Programming (SQP) optimization algorithm, the *MSQP* algorithm. The analysis in this section applies to modification of generic SQP, and in Section III these results are applied to dynamic optimization and NMPC.

We begin by defining the following general NLP:

$$\min_z \phi(z) \quad \text{s.t.} \quad g(z) = 0, \quad h(z) \geq 0. \tag{1}$$

where all functions are twice continuously differentiable. Assume that solutions exist and are bounded. We refer to the decision variables as *primal* variables.

Definition 1 (Regularity): A primal-point z is *regular* if and only if the Linear Independence Constraint Qualification (LICQ) [17] holds for g and h.

The Lagrangian is:

$$\mathcal{L}(z, \lambda, \mu) \triangleq \phi(z) + \lambda^\top g(z) + \mu^\top h(z), \tag{2}$$

where λ and μ are the Lagrangian multipliers—*dual variables*—corresponding to the equality and inequality constraints respectively. Given regularity, the Karush-Kuhn-Tucker (KKT) conditions are necessary for optimality:

$$\nabla_z \mathcal{L}(z, \lambda, \mu) = 0 \quad \text{(Stationarity)} \tag{3a}$$
$$g(z) = 0 \quad \text{(Primal feasibility)} \tag{3b}$$
$$h(z) \geq 0 \quad \text{(Primal feasibility)} \tag{3c}$$
$$\mu \leq 0 \quad \text{(Dual feasibility)} \tag{3d}$$
$$\mu \odot h(z) = 0 \quad \text{(Complementary).} \tag{3e}$$

An SQP algorithm is an iterative optimization algorithm that, from an initial guess, updates a primal-dual guess as:

$$z \leftarrow z + \alpha d, \quad (\lambda, \mu) \leftarrow (1 - \alpha)(\lambda, \mu) + \alpha(\lambda_{\text{QP}}, \mu_{\text{QP}}) \tag{4}$$

with a suitable step length α, in direction $(d, \lambda_{QP}, \mu_{QP})$ given as the primal-dual solution to the following QP-subproblem:

$$\min_d \quad \frac{1}{2} d^\top W(z, \lambda, \mu) d + J(z) d \tag{5a}$$
$$\text{s.t.} \quad g(z) + G(z) d = 0 \tag{5b}$$
$$h(z) + H(z) d \geq 0 \tag{5c}$$

The data W, J, G, H are defined as

$$\begin{aligned} W(\cdot) &\triangleq \mathcal{M}_W \left(d_z^2 \mathcal{L}(z, \lambda, \mu) \right) & J(z) &\triangleq \mathcal{M}_J \left(d_z \phi(z) \right) \\ G(z) &\triangleq \mathcal{M}_G \left(d_z g(z) \right) & H(z) &\triangleq \mathcal{M}_H \left(d_z h(z) \right), \end{aligned} \tag{6}$$

where $\mathcal{M}_{W,J,G,H}(\cdot) \equiv \text{id}(\cdot)$, i.e., the derivative information is unchanged, and d_z is the total derivative with respect to z. We now define the MSQP algorithm to be the algorithm above, except that the modification functions $\mathcal{M}_{W,J,G,H}$ need only satisfy: 1) W, J, G, H are locally bounded, and 2) $Z(z)^\top W(z, \lambda, \mu) Z(z) \succeq 0$, where Z is the null space of

$$\Gamma(z) \triangleq [G^\top(z), \ \hat{H}^\top(z)]^\top. \tag{7}$$

We make use of hat, $\hat{\cdot}$, to extract rows of H corresponding to the active set at a given iterate, and check, $\check{\cdot}$, for the complement.

A. Halting Points

SQP methods are KKT-solvers in nature, however, an MSQP algorithm does not guarantee a KKT point, let alone an optimum, because of the modified gradients. It follows that we forfeit the ability to rely on the KKT conditions as a means to understand and analyze the solution. In this section we will assume that the MSQP algorithm converges to a *'solution'*, and that this point is characterized by having a Newton step equal to zero. We then develop a basic understanding of these points by deriving a set of modified KKT conditions; *halting conditions*. We begin by defining the notion of a halting point:

Definition 2: In the context of an MSQP algorithm applied to an NLP, we say that a primal-point z° of that NLP is a "**halting point** under the dual-point $(\lambda^\circ, \mu^\circ)$" if and only if the modified QP-subproblem (5) generated at that primal-dual-point $(z^\circ, \lambda^\circ, \mu^\circ)$ has a global optimum at $(d, \lambda_{QP}, \mu_{QP}) = (0, \lambda^\circ, \mu^\circ)$.

We will use the term "halting point" for a primal-point z° if it is a halting point under at least one dual-point. The idea is that an SQP-type algorithm will not leave a halting point—*it halts*. We also define the notion of m-regularity, which ensures LICQ for modified QP-subproblems:

Definition 3 (m-Regularity): A primal-point z is *m-regular* if and only if $\Gamma(z)$ has full row rank.

In the context of an MSQP algorithm applied to (1), let the set of all m-regular halting points be denoted by $\mathbb{H}$.

Theorem 1 (Halting Conditions): Consider an MSQP algorithm applied to (1). Then, $z \in \mathbb{H}$ if and only if there exists a dual-point (λ, μ) such that

$$J^\top(z) + G^\top(z)\lambda + H^\top(z)\mu = 0 \quad \text{(m-Stationarity)} \tag{8a}$$
$$g(z) = 0 \quad \text{(Primal Feasibility)} \tag{8b}$$
$$h(z) \geq 0 \quad \text{(Primal Feasibility)} \tag{8c}$$
$$\mu \leq 0 \quad \text{(Dual Feasibility)} \tag{8d}$$
$$\mu \odot h(z) = 0 \quad \text{(Complementarity)} \tag{8e}$$

Proof: Under m-regularity, the KKT conditions of (5):

$$W(z)d + J^\top(z) + G^\top(z)\lambda + H^\top(z)\mu = 0, \tag{9a}$$
$$g(z) + G(z)d = 0, \tag{9b}$$
$$h(z) + H(z)d \geq 0, \tag{9c}$$
$$\mu \leq 0, \tag{9d}$$
$$\mu \odot (h(z) + Hd) = 0, \tag{9e}$$

are necessary. Since, by construction, we require $Z^\top W Z \succeq 0$, (9) is also sufficient. Therefore, stating that an m-regular point z is a halting point is equivalent to stating that there exists a dual-point such that (z, λ, μ) satisfies (9) with $d = 0$. By inserting $d = 0$ into (9) we recover (8), proving the equivalence:

$$z \in \mathbb{H} \iff \exists(\lambda, \mu) : (z, \lambda, \mu) \text{ satisfies (8).} \tag{10}$$

■

Corollary 1 (Feasibility and Consistent Active Set): Let $z^\circ \in \mathbb{H}$, then z° is feasible with respect to (1). Furthermore, at z°, the active set of the QP-subproblem (5) with $d = 0$, is the same as the active set of the original NLP (1) at z°.

Proof: At $z^\circ \in \mathbb{H}$, Theorem 1 says that (8b) and (8c) hold, which amount to both the original constraints of (1), and the constraints of (5) with $d = 0$. ■

Remark 1 (Hessian Modification Independence): The halting conditions (8) are entirely independent of the modified hessian $W(z, \lambda, \mu)$, thus by Theorem 1, so is the set of all m-regular halting points, $\mathbb{H}$.

Remark 2 (Structural Similarity to KKT conditions): The halting conditions (8) are identical in structure to the KKT conditions (3), the only difference being the modified stationarity condition. Therefore, up to reasonable choices of modifications (6), the assumption $\mathbb{H} \neq \emptyset$ is not *structurally* stronger than existence of KKT points.

B. Problem Re-Formulation

Because of the equivalence in Theorem 1, we may neglect the application of a conceptual MSQP algorithm and instead reformulate (1) by appending the modified stationarity conditions (8a),(8d),(8e):

$$\begin{aligned} \min_{z,\lambda,\mu} \quad & \phi(z) \\ \text{s.t.} \quad & 0 = g(z), \\ & 0 \leq h(z) \perp \mu \leq 0, \\ & 0 = J^\top(z) + G^\top(z)\lambda + H^\top(z)\mu, \end{aligned} \tag{11}$$

The constraints of problem (11) coincide exactly with the halting conditions (8). Thus, the feasible region is equal to the set of halting points, from which one should choose the candidate with the smallest objective. This reformulation amounts to Mathematical Programs with Complementarity Constraints (MPCC), which generally require special solvers [18]. As a theoretical tool, (11) places the OSM strategy within established optimization theory and provides a basis for subsequent analysis of the OSM-NMPC extension.

III. Output Sensitivity Modification for Dynamic Optimization

Building on the idea of modifying the gradients of an SQP algorithm, we propose an *output sensitivity modification (OSM)* strategy for dynamic optimization problems, which in turn induces the OSM-NMPC scheme.

Consider the discrete-time system

$$x_{k+1} = f_d(x_k, u_k), \tag{12a}$$
$$y_k = \gamma(x_k). \tag{12b}$$

where $x_k \in \mathbb{R}^{n_x}$, $u_k \in \mathbb{R}^{n_u}$, $y_k \in \mathbb{R}^{n_y}$ are the state, input and output vectors at time k, and $f_d : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \to \mathbb{R}^{n_x}$ and $\gamma : \mathbb{R}^{n_x} \mapsto \mathbb{R}^{n_y}$ are sufficiently smooth. We consider a horizon length of $N \in \mathbb{N}$, and define the collection of input, state, and output vectors: $u \triangleq (u_0, u_1, \ldots, u_{N-1})$, $x \triangleq (x_1, x_2, \ldots, x_N)$, and $y \triangleq (y_1, y_2, \ldots, y_N)$. Furthermore, consider a cost function:

$$\tilde{\phi}(u, y) = \sum_{k=0}^{N-1} \ell_k(u_k, y_{k+1}) \tag{13}$$

The dynamic optimization problem is then formulated as (1), with $\phi = \tilde{\phi}$, where the decision vector is $z^\top \triangleq [u^\top, \bar{z}^\top]$ and $\bar{z}$ collects remaining decision variables.

A. The OSM Strategy

Once the dynamic optimization problem is formulated, the OSM strategy conceptually modifies the gradients of an SQP algorithm applied to that problem. In doing so, we accept not necessarily finding the optimum, in exchange for obtaining another solution—*a halting point*—with some desired properties. The idea is that during search, the optimizer does not 'perceive' the true sensitivity of the predicted output trajectory with respect to the current input trajectory. Instead, it can only account for the modified sensitivities, yielding a solution in which the preferred sensitivity structure is imposed.

Consider the input-output sensitivity:

$$\mathcal{D} \triangleq \frac{dy}{du} = \begin{bmatrix} \mathcal{D}_{1,0} & & \\ \vdots & \ddots & \\ \mathcal{D}_{N,0} & \cdots & \mathcal{D}_{N,N-1} \end{bmatrix}, \tag{14}$$

where

$$\mathcal{D}_{k,l} \triangleq \frac{dy_k}{du_l} \in \mathbb{R}^{n_y \times n_u}. \tag{15}$$

The (total) Jacobians with respect to the input trajectory can then be expressed as follows:

$$d_u(\tilde{\phi}, g, h) = \partial_u(\tilde{\phi}, g, h) + \partial_y(\tilde{\phi}, g, h)\mathcal{D} \tag{16}$$

The OSM strategy then consists of modifying the output sensitivity matrix before applying it in (16). Denote the modified sensitivity by $\bar{\mathcal{D}}$, and consider the following modified gradients:

$$(\bar{J}_u, \bar{G}_u, \bar{H}_u) = \partial_u(\tilde{\phi}, g, h) + \partial_y(\tilde{\phi}, g, h)\bar{\mathcal{D}} \tag{17}$$

The complete modified gradients are

$$\bar{J} \triangleq \begin{bmatrix} \bar{J}_u & \frac{d\tilde{\phi}}{d\bar{z}} \end{bmatrix}, \bar{G} \triangleq \begin{bmatrix} \bar{G}_u & \frac{dg}{d\bar{z}} \end{bmatrix}, \bar{H} \triangleq \begin{bmatrix} \bar{H}_u & \frac{dh}{d\bar{z}} \end{bmatrix}. \tag{18}$$

These gradients ultimately define the OSM strategy via the halting conditions (8), rather than the application of MSQP. Note that the modified sensitivity matrix applied in (17) does not need to be the same for all gradients $\bar{J}$, $\bar{G}$, and $\bar{H}$. This remains a design choice. Notably, if the modifications limit control freedom—such as a decoupling strategy—one may want to modify only the objective gradient, and choose $\bar{G} = \frac{dg}{dz}$, $\bar{H} = \frac{dh}{dz}$, such that one does not restrict control freedom with respect to satisfying constraints. This is illustrated in Section IV. Lastly, the OSM-NMPC scheme is defined by the first input u_0 of the OSM solution.

To complete the OSM strategy, one must choose the modification strategy itself, that is, *how should the sensitivity matrix be modified*: $\mathcal{D} \mapsto \bar{\mathcal{D}}$? For concision, we will not discuss various ways the sensitivity could meaningfully be modified. Instead, we consider only *input-output decoupling*, in which certain inputs are desired decoupled from certain outputs from a control perspective. To this end, we set the corresponding elements of the sensitivity matrix to zero.

IV. Numerical Example: Exothermic CSTR

As a numerical example, we consider a simple benchmark process system: an exothermic Continuously Stirred Tank Reactor (CSTR) with a single first-order, irreversible reaction

$$A \to B,$$

in liquid phase (see Example 2.5 in [19]). The example is a good benchmark because it is simple enough to retain intuition about the dynamic mechanisms, such that the solutions remain interpretable, yet inhabits some of the challenging nonlinear dynamic interplay in real-world process systems. The process is modeled as a nonisothermal system, where mass- and energy balances are strongly coupled, and the admissible control signals are limited by the cooling capacity of the cooling jacket.

The aim is to concisely demonstrate decoupling effects via the proposed OSM-NMPC scheme. Therefore, we do not elaborate on the necessity of decoupling itself, nor other sensitivity modifications, which is already established in the introduction.

Consider:

State vector: $x = [C_A, T]^\top$
C_A concentration of reactant A [mol/L]
T reactor temperature [K]

Input vector: $u = [q, T_c]^\top$
q inlet volumetric flow rate [L/s]
T_c coolant (jacket) temperature [K]

Output vector: $y = T$

The dynamics are

$$\dot{C}_A = \frac{q}{V}\big(C_{Af} - C_A\big) - k(T)C_A, \tag{19a}$$

$$\dot{T} = \frac{q}{V}\big(T_f - T\big) + \frac{\Delta H_r}{\rho C_p}\, k(T)C_A - \frac{UA}{\rho C_p V}\big(T - T_c\big), \tag{19b}$$

with Arrhenius rate law $k(T) = k_0 \exp\Big(-\frac{E}{T}\Big)$. The reactant feed temperature and concentration, reactor volume, fluid density and heat capacity are considered constant, and all parameters are summarized in Table I. The study consists of two examples: 1) one that clearly highlights the decoupling effect, and 2) another that demonstrates that we retain feasibility with respect to the full nonlinear model. In both examples we seek to regulate the system to a nominal high-temperature, unstable, steady state reference point, close to thermal runaway:

TABLE I
CSTR Model Parameter Values

Parameter	Value	Unit	Description
V	100	L	reactor volume
ρ	1	kg/L	liquid density in the reactor
C_p	0.239	kJ/(kg · K)	heat capacity of reactor fluid
UA	$0.833\bar{3}$	kJ/(s · K)	overall heat transfer coefficient
C_{Af}	1	mol/L	inlet concentration of reactant A
T_f	295	K	inlet feed temperature
k_0	1.2×10^9	1/s	Arrhenius pre-exponential factor
E	8750	K	act.energy over gas constant (E_a/R)
ΔH_r	50	kJ/mol	heat of (exothermic) reaction

$$x_r = \begin{bmatrix} C_{A,r} \\ T_r \end{bmatrix} = \begin{bmatrix} 0.31008 \\ 360 \end{bmatrix}, u_r = \begin{bmatrix} q_r \\ T_{c,r} \end{bmatrix} = \begin{bmatrix} 1.5 \\ 325.870 \end{bmatrix}.$$

Furthermore, the reactor temperature may not exceed $361K$, and the minimum cooling jacket temperature is $315K$.

The objective is expressed via quadratic stage costs as:

$$\ell(y_{k+1}, u_k) = \frac{100}{T_r}\Delta T_{k+1}^2 + \Delta u_k^\top \begin{bmatrix} \frac{1}{q_r} & 0 \\ 0 & \frac{10}{T_{c,r}} \end{bmatrix} \Delta u_k, \tag{20}$$

where $\Delta T_k = (T_k - T_r)$, and $\Delta u_k = (u_k - u_r)$. The two case studies then consist of recovering from additive disturbances $x_{d,1} = [0, -2]^\top$ and $x_{d,2} = [0.1, -1]^\top$.

We employ a standard finite-horizon NMPC formulation. The continuous-time model (19) is discretized with a fixed-step explicit RK4 scheme, resulting in a discrete-time map; $x_{k+1} = F(x_k, u_k)$, with a prediction horizon of length $N = 30$. Given an initial state x_{init}, we solve:

$$\min_u \quad \sum_{k=0}^{N-1} \ell(y_{k+1}, u_k) \tag{21a}$$

$$\text{s.t.} \quad x_0 = x_{\text{init}}, \tag{21b}$$

$$x_{k+1} = F(x_k, u_k), \qquad k \in \mathbb{Z}_{0,N-1} \tag{21c}$$

$$y_k = \gamma(x_k) \qquad k \in \mathbb{Z}_{1,N} \tag{21d}$$

$$T_k \le 361\ [K], \qquad k \in \mathbb{Z}_{1,N} \tag{21e}$$

$$T_{c,k} \ge 315\ [K], \qquad k \in \mathbb{Z}_{0,N-1}. \tag{21f}$$

A. Closed-Loop Simulation

The case study makes use of a common industrial input-output pairing decision: the jacket temperature T_c is the primary control input used to regulate the reactor temperature T. The inlet flow q should therefore only be utilized if necessary to satisfy constraints. To this end, we may design an OSM strategy that masks the sensitivity matrix in the following way: $\frac{\partial T_i}{\partial q_j} = 0$ for all $(i, j) \in \mathbb{Z}_{1,N} \times \mathbb{Z}_{0,N-1}$. This structural modification encourages the optimizer to use exclusively T_c for temperature regulation whenever feasible. In this example

we choose to modify only the objective gradient $\bar{J}$ (denoted *"objective modification"*), yet elaborate on the effect of also modifying the constraint gradients $\bar{G}, \bar{H}$ (*"full modification"*).

Both examples are simulated in closed-loop using both the classical NMPC formulation and OSM-NMPC, the latter employing the described decoupling strategy. The simulation starts at the disturbed state $x_r + x_{d,\{1,2\}}$, and simulates $2min$ forwards in time. The horizon length in each example is $1min$. The results are shown in Fig. 1 and Fig. 2.

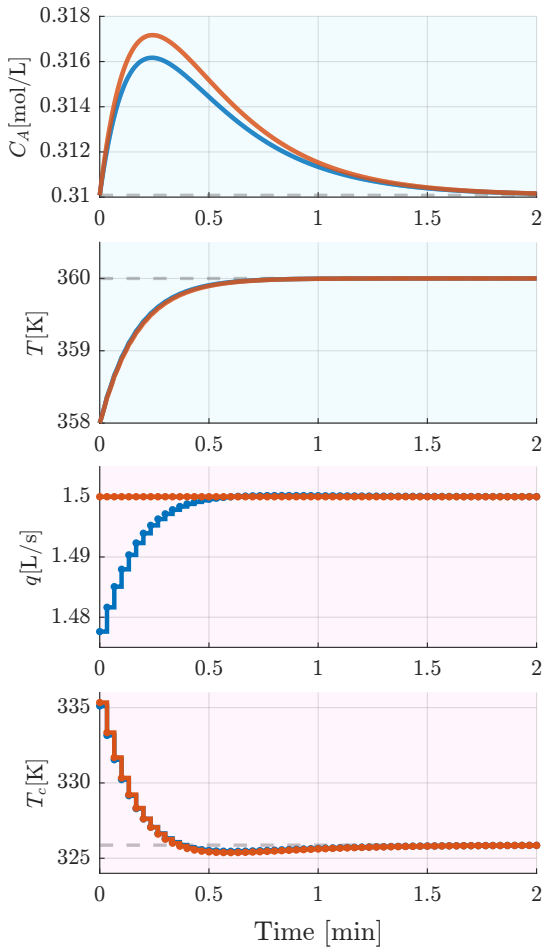

Fig. 1. (Example 1) Closed-loop solutions of (19) under classical NMPC (blue) and OSM-NMPC (red) with the described decoupling approach. Background colors: blue-states, pink-inputs. Gray dashed line: nominal values. The sampling time and the integration step size of the predictions are both $\boldsymbol{\Delta t = 2s}$

Example 1 (decoupling effect): From Fig. 1, we see that the temperature is being corrected by use of the cooling jacket temperature in both classical NMPC (blue) and OSM-NMPC (red). The classical NMPC controller assists this maneuver by reducing the feed flow, which is cooler than the reactor, thus reducing its cooling effect. The OSM controller, on the other hand, does not 'perceive' this effect, as it has been eliminated from the linearized sensitivity matrix, and subsequently does not aid the maneuver by reducing flow. Instead, the flow rate is kept at its nominal value, which is the behavior we sought. In this case, no constraints are active, meaning that any modification of the constraint gradients would not affect the solution. Hence *objective-only* and *full* modification are equivalent in this case.

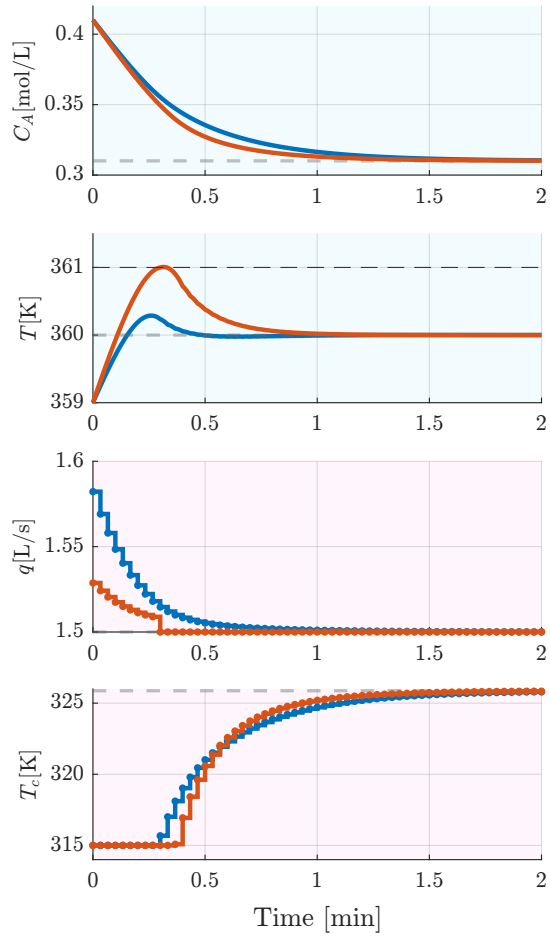

Fig. 2. (Example 2) Red dashed: constraint. See caption of Fig. 1.

Example 2 (coupled feasibility): In Fig. 2 we see that the reactant feed rate is used extensively by the classical NMPC controller to cool the reactor, preventing thermal runaway, even at the cost of a slightly increased concentration—which has a smaller and less immediate effect on the temperature. Upon eliminating the cooling effect of the inlet flow from the controller's 'perspective', the controller can no longer determine that the increasing flow rate aids the maneuver from an optimality point of view. Nevertheless, the flow rate is also used to an extent by the OSM-NMPC controller (red) in this example. The cooling capacity is clearly saturated, causing thermal runaway—or at least constraint violation $T > 361K$—if the flow rate is idle. Therefore, as stated in Corollary 1: *halting points are feasible*, the controller must use the cooling effect of the inlet flow to attain feasibility. We remark that this effect appears only under *objective-only* modification. The same problem, using *full* modification, turns out to be infeasible due to lack of control freedom when removing the $q \to T$ coupling from the constraint gradient. The full modification for OSM is analogous to the linear MPC MV-CV elimination strategy, in which the model is modified for all evaluations in the related QP. Hence, this example highlights an advantage of our method under objective-only modification over preexisting methods.

V. Concluding Remarks

In this paper, we conceptualized a modified-gradient SQP algorithm, in which the gradient information is modified before generating a Newton step. The results we present are centered around the notion of halting points of the MSQP algorithm, which is to say that we are only concerned with the *'solutions'* of the algorithm and the resulting feedback law. We proposed the reformulation (11), which amounts to directly solving the halting conditions, but cast as a nonlinear program with complementarity constraints, i.e., an MPCC. This viewpoint frames the OSM-NMPC controller as an MPCC-based NMPC controller, which facilitates future stability and robustness analysis. The results in Section IV were computed by solving the halting conditions via an exact Fischer-Burmeister reformulation of complementarity [20], [21]. It is also worth emphasizing that upon testing the MSQP algorithm itself, we found that it converges quickly to the same halting points from close initial guesses, but exhibits poor convergence in general.

In future work, it would be interesting to develop strategies that allow us to employ MSQP directly. Such strategies would be attractive because they are independent of the computation—and even existence—of modification function derivatives. Such strategies should involve a globalization strategy via tailored merit functions and Hessian design, and potentially additional modification of the gradients, to provide convergence guarantees. Future work will also explore conditions necessary for existence of halting points and bounds on sub-optimality, and the efficacy of OSM in more complex system settings. Furthermore, we are currently developing the equivalent strategy for *multiple shooting* and *direct collocation* formulations.

Another question that arises: *could one instead simply modify the objective function?* For 'most' sensitivity modifications, the resulting gradients do not constitute conservative vector fields, meaning that we cannot find a scalar function whose gradient is the modified gradient from OSM. Nevertheless, this does not rule out the existence of a function whose gradient is a sufficient approximation, and is therefore useful. Exploring this idea further is of interest.

A last remark on the impact of the feasibility result (Corollary 1): For many systems, satisfying feasibility is a prerequisite, with little wiggle room. The OSM strategy allows us to alter the dynamics from an optimality perspective, while enforcing constraints with respect to the true model dynamics. When only the objective gradient is modified, this separation causes the controller to nominally neglect the true dynamics, yet whenever feasibility becomes a limiting factor, it 'automatically activates' the true model dynamics. This is a powerful observation since it allows us to impose structural preferences without compromising feasibility, which is often safety-critical. Note that this reactivation requires no additional implementation of logical statements.

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